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Essay 2

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Fluency that Lacks Creativity: Generative AI and the Limits of Machine-Made Music

Imagine you are on Suno, where it says “Enter to prompt create your music”, and then you have a friend who is obsessed with Cantopop and she types “Generate me a melancholic Cantopop ballad”. Surely, Suno can generate that ballad within a couple of seconds, but your friend hates it and you start to analyse why. That all comes down to realising the origin of Suno which is heavily based in Machine Learning. While ML is being used to sort and label and now even generates, there is in fact a substantial technical gap beyond classification and that fluency is not equivalent to creativity defined from a human perspective.

In fact, that is exactly the use of generative AI, which produces impressively novel-sounding outputs through sampling learned distributions. Yet, its "creativity" is bounded by training data in ways that expose both a cultural gap and a deeper philosophical one about what music creation actually is. Therefore, what does generative AI actually do? There are two types of ML. Traditional ML involves classification and clustering such as figuring out patterns, grouping similar data points. A common example we encounter regularly is Spotify's recommendation system or YouTube's playlist system. Genre labelling also falls here but nothing novel is produced. The leap comes with generative models, which learn through a compressed representation of music through what is called a latent space, and then sample from it to produce outputs that were not in the training set. According to Roberts et al. (2018), MusicVAE encodes melodies into a latent vector and can interpolate between two melodies to generate a third that is

genuinely neither. This is the core idea: the model does not retrieve or copy. It samples from a learned space of musical possibility. Also, Jukebox (Dhariwal et al., 2020) extends this further. While MusicVAE operates on symbolic MIDI, Jukebox operates on raw audio, using a VQ-VAE to compress audio into discrete tokens and then an autoregressive transformer to predict what comes next. This enables coherent music with vocals to be produced within minutes, whereas traditional music production requires hours or even days. The truth is that these systems do produce outputs that never existed before. The question is not whether the output is new but rather whether the process itself is creative enough.

Personally, I believe nothing from AI-generated music has a genuine origin or can be deemed creative. While people may counterargue that humans also learn from existing music, human music creation is much more nuanced and based on induction from lived experience. A human composer internalises music through a series of embodied, cultural, emotional, and physical experiences. When a person hears a song at a funeral, it changes how they write, and vice versa at a wedding. That context is generative and it is not in any dataset. AI samples from a learned distribution. It cannot generate meaningfully from outside that distribution. When an AI output sounds surprising, it is because the latent space is high-dimensional and we cannot predict where the sample lands. That is not intuition. Instead, that is called variance. Minsky's *Society of Mind* (1986) highlights that creativity in humans emerges from competing internal agents with different goals, memories, and emotions. There is no analog to that in a VAE.

Considering the Cantopop case as an example makes the architectural limitation of AI-generated music concrete in a way that is hard to dismiss. Cantonese is a tonal language with six or nine lexical tones, where pitch is used to distinguish word meaning rather than simply convey emotional emphasis as in English. This is not a stylistic feature, but rather a phonological

constraint that governs how melody and lyrics must relate. Research by Chan (1987) indicated that in Cantopop, lexical tones must be maintained consistently across stanzas within a song. Li, Carter-Enyi, and Aina (2024) confirm that the direction of melodic intervals must generally mirror the direction of tonal target transitions, a relationship called tone-tune mapping. A rising tonal sequence must be set to a rising melodic sequence. Setting a high-to-low tone pair on a rising melody produces something that native Cantonese speakers immediately recognise as wrong, not just unusual. The melody makes the words mean something different, or makes them unintelligible entirely. No current generative model has a mechanism to enforce this constraint, because it was never in the training data to begin with. According to the "Missing Melodies" paper (Mehta, Chauhan, and Choudhury, 2024), AI music generation is overwhelmingly trained on music from the Global North, with approximately 86% of total dataset hours and over 93% of researchers focused on that corpus. Genres from the Global South account for only 14.6% of the data. Cantopop is a fraction of that fraction. A model trained overwhelmingly on Western pop and classical music does not learn tone-tune mapping because Western pop and classical music do not have it. The model's creative range is literally the shape of its training corpus and is heavily biased. The output may sound like Cantopop in a shallow sense, using familiar instrumentation or chord progressions learned from adjacent genres, but it is structurally incorrect at the level of linguistic grammar and pronunciation. That is not a gap where more data will easily close, it is an architectural one.

Additionally, it is known that Jukebox, MusicLM (Agostinelli et al., 2023), Suno, and Udio were all trained on massive corpora of human-made music, most of it without consent. In 2024, the Recording Industry Association of America filed lawsuits on behalf of Universal, Sony, and Warner against Suno and Udio for training on copyrighted material without

authorisation. Both companies admitted making use of copyrighted recordings and argued fair use as their defence. The authorship question remains open: if the model's generative capacity is entirely derived from artists who never agreed to participate in the first place, who owns the output? The company? The user? The original artists, who are uncredited? Even setting the legal question aside, there is a philosophical one. If creativity requires originality, and originality requires not being wholly derived from prior work without acknowledgment, then AI-generated music has a problem at its foundation that adding more licensed data does not resolve but instead creates more controversial problems.

While generative AI is a real advance that goes beyond classification and produces outputs that are new in a technical sense, its novelty is not creativity. Creativity involves generating from a position of embodied experience, cultural awareness, and emotional stake. AI has none of these. The Cantopop case makes this especially evident when melody is co-constrained by language, culture, and tone in ways that are not captured by any current model architecture. The output is not just stylistically thin but structurally incorrect and unoriginal in the deepest sense. The question going forward is not whether AI can generate music, because clearly it can. The bigger question is what humans lose when we stop asking who is generating it and the ultimate purpose of music creation.

Citation

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